Rex Tin Long Fung

E-mail: rexfung@umich.edu
Website: https://rextlfung.github.io

EDUCATION

University of Michigan, Ann Arbor, MI — Sep 2022 - Sep 2027

PhD Candidate in Biomedical Engineering specializing in computational imaging: co-designing sensing hardware (MRI acquisition) with learned and model-based reconstruction to push imaging into hard regimes (high resolution, heavy undersampling), with applications to accelerating functional MRI.

University of Michigan, Ann Arbor, MI — Sep 2022 - Aug 2024

Program: MS, Biomedical Engineering (Imaging and Ultrasonics) **GPA**: 3.65/4.00

Relevant Coursework: Machine Learning, Optimization, Image Processing, Medical Imaging Systems, Linear Algebra, Probability

University of Michigan, Ann Arbor, MI — Sep 2018 - May 2022

Program: BS, Neuroscience major, Electrical Engineering minor **GPA**: 3.78/4.00

Relevant Coursework: Digital Signal Processing, Electrical Biophysics, Embedded Control Systems, Sensing and ML for Neural Interfaces

RESEARCH EXPERIENCE

Functional MRI Lab, University of Michigan — Sep 2022 - Present

Graduate Student Research Assistant — *Advisors: Dr. Jon-Fredrik Nielsen, Dr. Douglas Noll*

- Co-designed acquisition and reconstruction for a novel randomly undersampled (Poisson-disc) 3D-EPI fMRI sensing system, achieving 10× acceleration and sub-2 mm resolution, a regime where conventional acquisition fails.
- Developed a custom 4D iterative reconstruction in Julia and Python that models multiscale low-rank and sparse signal structure, reconstructing 100 GB+ datasets on GPUs/HPC clusters and improving the accuracy of functional activation maps.
- Prototyped sequences on research MRI scanners under real hardware constraints (gradient slew, timing, SAR, scan time), iterating from simulation to phantom to human experiments.
- Built open-source, vendor-agnostic Pulseq sequences used by 7+ international institutions (e.g., Harvard, Freiburg), collaborating across physics, engineering and neuroscience groups.

Bo Duan Lab, University of Michigan — Jul 2020 - May 2022

Undergraduate Research Assistant — *Advisor: Dr. Bo Duan*

- Cut data analysis time by 90% by building OpenCV computer-vision pipelines for behavioral tracking in 30+ hours of video and automated cell counting in 200+ microscopy images.
- Co-authored 1 journal article in *Nature Communications*, characterizing a dedicated skin-to-brain neural circuit for cool sensation in mice.

TECHNICAL SKILLS

Imaging & Machine Learning: Computational Imaging, Inverse Problems, Convex/Iterative Optimization, Machine Learning, Computer Vision, Signal & Image Processing

Programming: Python (PyTorch, NumPy, OpenCV), Julia, MATLAB, GPU programming, Unix/Linux, HPC clusters, Git/GitHub, LaTeX

Systems & Research: MRI scanner hardware and pulse sequence programming, Experimental Design, Academic Writing and Presentations

Languages: English, Cantonese, Mandarin Chinese, Japanese

Rex Tin Long Fung

E-mail: rexfung@umich.edu
Website: https://rextlfung.github.io

JOURNAL PUBLICATIONS

[1] R.A. Lobos, X. Wang, **R.T.L. Fung**, Y. He, D. Frey, D. Gupta, Z. Liu, J.A. Fessler, D.C. Noll. "Spatiotemporal Maps for Dynamic MRI Reconstruction.", *IEEE Transactions on Computational Imaging* (2026). DOI.

[2] H. Lee, C.C. Hor, L.R. Horwitz, A. Xiong, X.Y. Su, D.R. Soden, S. Yang, W. Cai, W. Zhang, C. Li, C. Radcliff, A. Dinh, **T.L.R. Fung**, I. Rovcanin, K.P. Pipe, X.Z.S. Xu, B. Duan. "A dedicated skin-to-brain circuit for cool sensation in mice.", *Nature Communications* (2025). DOI.

MANUSCRIPTS UNDER REVISION

[R1] **R.T.L. Fung**, R.A. Lobos, J.A. Fessler, D.C. Noll, J.-F. Nielsen. "ArbEPI: 3D-EPI sequences from arbitrary 2D sampling masks, with multiscale low-rank reconstruction for fMRI.", *Magnetic Resonance in Medicine*, under revision.

CONFERENCE POSTERS

[1] **R.T.L. Fung**, D.C. Noll, J.-F. Nielsen. "ArbEPI: A Framework for Generating 3D-EPI Sequences from Arbitrary Sampling Patterns" *In Vivo Magnetic Resonance, Gordon Research Conferences*, 2026. 20 minute talk and traditional poster.

[2] **R.T.L. Fung**, R.A. Lobos, J.A. Fessler, D.C. Noll, J.-F. Nielsen. "Sub 2 mm Resolution fMRI at 3T using Randomly Undersampled 3D-EPI with Locally Low-Rank + Temporally Sparse Reconstruction" *International Society for Magnetic Resonance in Medicine Annual Meeting*, 2025. Digital poster #3196.

[3] S. Peltier, M. Egan, **R. Fung**, Q. Chen, M. Zaitsev, J.-F. Nielsen. "Harmonization of an SMS-EPI fMRI protocol using Pulseq: Sequence implementation and ABCD QC metrics." *Organization for Human Brain Mapping Annual Meeting*, 2024. Digital poster #3804.

[4] Y. He, **R. Fung**, J.-F. Nielsen. "High-Accuracy Ultra-short Inner-Volume Saturation Pulse for 3D Steady-State Imaging." *International Society for Magnetic Resonance in Medicine Annual Meeting*, 2024. Digital poster #3249.

PROJECTS

Volleybot, github.com/rextlfung/volleybot — Personal Project

Built a YOLOv8 (PyTorch) ball-tracking pipeline for fixed-camera gym video, fine-tuned on $\sim$500 labeled frames to beat a pretrained YOLOv8x baseline by +17pp detection rate at 6$\times$ the speed; added a rally state machine and live/dead-play classifier to auto-cut highlight reels.

TEACHING EXPERIENCE

BME/ECE 516 Medical Imaging Systems, University of Michigan — Sep 2024 - Dec 2024

- Mentored 20-30 students in Medical Imaging Systems (MRI, ultrasound) through weekly office hours and Piazza.
- Released and graded homework and the final exam on Canvas over 15 weeks.

HONORS AND AWARDS

James B. Angell Scholar, University of Michigan — Mar 2021

Students who achieve an "A" record (all grades of A+, A, or A-) for two or more consecutive terms